

12. This time, drag **Engine size** to Rows and **Horsepower** to the **Values** field.



Car sales data



File

Edit

View

Insert

Format

Data

Tools

Extensions

Help



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123

Defaul...



A1



fx

Engine\_size

|    | A           | B                 | C | D | E |
|----|-------------|-------------------|---|---|---|
| 1  | Engine_size | SUM of Horsepower |   |   |   |
| 2  |             | 0                 |   |   |   |
| 3  | 1           | 55                |   |   |   |
| 4  | 1.5         | 92                |   |   |   |
| 5  | 1.6         | 106               |   |   |   |
| 6  | 1.8         | 1059              |   |   |   |
| 7  | 1.9         | 644               |   |   |   |
| 8  | 2           | 2168              |   |   |   |
| 9  | 2.2         | 522               |   |   |   |
| 10 | 2.3         | 1101              |   |   |   |
| 11 | 2.4         | 1680              |   |   |   |
| 12 | 2.5         | 1691              |   |   |   |

|    |     |      |
|----|-----|------|
| 13 | 2.7 | 769  |
| 14 | 2.8 | 586  |
| 15 | 2.9 | 201  |
| 16 | 3   | 2740 |
| 17 | 3.1 | 670  |
| 18 | 3.2 | 1306 |
| 19 | 3.3 | 850  |
| 20 | 3.4 | 1505 |

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Sheet1
Sheet2
Pivot Table 4
Pivot Table

13. Click on the arrow under the **Values** field and select **Average**.



Car sales data



File Edit View Insert Format Data Tools Extensions Help



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Default... ▾

A1 ▾

fx

Engine\_size

|    | A           | B              | C | D | E |
|----|-------------|----------------|---|---|---|
| 1  | Engine_size | SUM of Horsepo |   |   |   |
| 2  |             | 0              |   |   |   |
| 3  | 1           | 55             |   |   |   |
| 4  | 1.5         | 92             |   |   |   |
| 5  | 1.6         | 106            |   |   |   |
| 6  | 1.8         | 1059           |   |   |   |
| 7  | 1.9         | 644            |   |   |   |
| 8  | 2           | 2168           |   |   |   |
| 9  | 2.2         | 522            |   |   |   |
| 10 | 2.3         | 1101           |   |   |   |
| 11 | 2.4         | 1680           |   |   |   |
| 12 | 2.5         | 1691           |   |   |   |

|    |     |      |
|----|-----|------|
| 13 | 2.7 | 769  |
| 14 | 2.8 | 586  |
| 15 | 2.9 | 201  |
| 16 | 3   | 2740 |
| 17 | 3.1 | 670  |
| 18 | 3.2 | 1306 |
| 19 | 3.3 | 850  |
| 20 | 3.4 | 1505 |



+
≡
Sheet1 ▼
Sheet2 ▼
Pivot Table 4 ▼
Pivot Table

14. The pivot table is ready. You can also change the name of this table.

Car sales data

File Edit View Insert Format Data Tools Extensions Help

100% 123 Default... 10 B I

|    | A           | B             | C | D | E | F | G |
|----|-------------|---------------|---|---|---|---|---|
| 1  | Engine_size | AVERAGE of Ho |   |   |   |   |   |
| 2  |             | #DIV/0!       |   |   |   |   |   |
| 3  | 1           | 55            |   |   |   |   |   |
| 4  | 1.5         | 92            |   |   |   |   |   |
| 5  | 1.6         | 106           |   |   |   |   |   |
| 6  | 1.8         | 132.375       |   |   |   |   |   |
| 7  | 1.9         | 128.8         |   |   |   |   |   |
| 8  | 2           | 127.5294118   |   |   |   |   |   |
| 9  | 2.2         | 130.5         |   |   |   |   |   |
| 10 | 2.3         | 183.5         |   |   |   |   |   |
| 11 | 2.4         | 152.7272727   |   |   |   |   |   |
| 12 | 2.5         | 153.7272727   |   |   |   |   |   |
| 13 | 2.7         | 192.25        |   |   |   |   |   |
| 14 | 2.8         | 195.3333333   |   |   |   |   |   |
| 15 | 2.9         | 201           |   |   |   |   |   |
| 16 | 3           | 195.7142857   |   |   |   |   |   |
| 17 | 3.1         | 167.5         |   |   |   |   |   |
| 18 | 3.2         | 217.6666667   |   |   |   |   |   |
| 19 | 3.3         | 170           |   |   |   |   |   |
| 20 | 3.4         | 215           |   |   |   |   |   |

Sheet1 Sheet2 Pivot Table 4 Pivot Table 3 Pivot Table 1

#### Step 4: Visualize data using charts

Based on the output received from ChatGPT, after generating the pivot tables, you need to visualize your data using charts.

1. Let's create a few charts for your sales dashboard. Go to the sheet Pivot Table 1. Highlight the pivot table data for total sales by manufacturer. Click on **Insert** and then click on **Charts**.

Car sales data

File Edit View **Insert** Format Data Tools Extensions Help

A1:B32 | fx Manufa

|    | A            |     |
|----|--------------|-----|
| 1  | Manufacturer | SUM |
| 2  | Acura        |     |
| 3  | Audi         |     |
| 4  | BMW          |     |
| 5  | Buick        |     |
| 6  | Cadillac     |     |
| 7  | Chevrolet    |     |
| 8  | Chrysler     |     |
| 9  | Dodge        |     |
| 10 | Ford         |     |
| 11 | Honda        |     |
| 12 | Hyundai      |     |
| 13 | Infiniti     |     |

- Cells
- Rows
- Columns
- Sheet
- Chart**
- Pivot table
- Image
- Drawing
- Function
- Link

Shift+F11

Ctrl+K

|                                                            |            |                                              |
|------------------------------------------------------------|------------|----------------------------------------------|
| 13                                                         | mini       |                                              |
| 14                                                         | Jaguar     | <input checked="" type="checkbox"/> Checkbox |
| 15                                                         | Jeep       |                                              |
| 16                                                         | Lexus      | <input type="checkbox"/> Comment Ctrl+Alt+M  |
| 17                                                         | Lincoln    |                                              |
| 18                                                         | Mercedes-B | <input type="checkbox"/> Note Shift+F2       |
| 19                                                         | Mercury    | 237.555                                      |
| 20                                                         | Suzuki     | 180.895                                      |
|                                                            |            |                                              |
| <div> + ≡ Sheet1 Sheet2 Pivot Table 4 Pivot Table 3 </div> |            |                                              |

2. Select **Column chart** under the **Chart type**, and your chart is ready.



Car sales data



File Edit View Insert Format Data Tools Extensions Help



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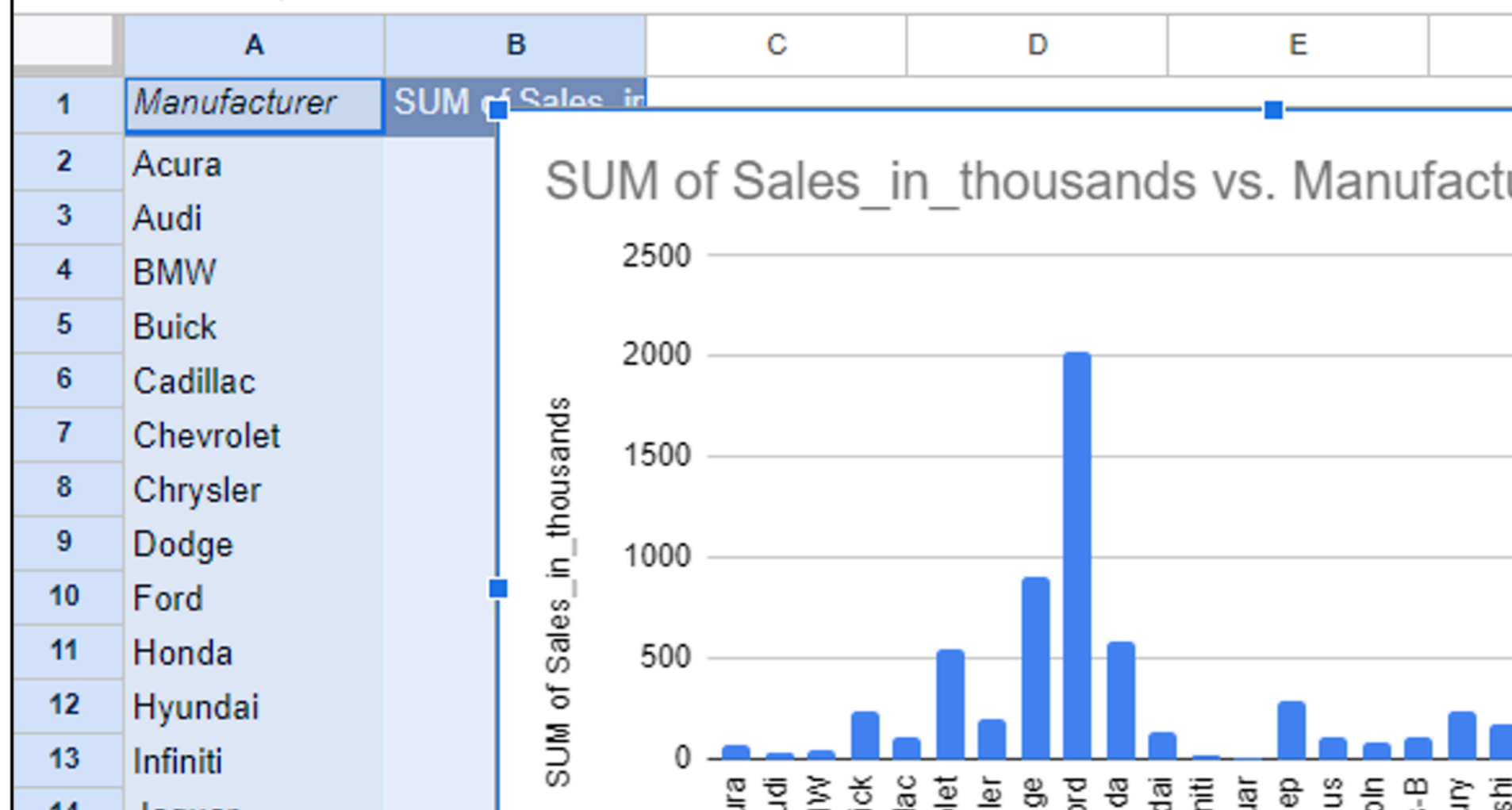
123

Defaul... ▾



A1:B31 ▾

*fx* Manufacturer







Car sales data



File Edit View **Insert** Format Data Tools Extensions Help



Cells



Rows



Columns



Sheet

Shift+F11



Chart



Pivot table



Image



Drawing



Function



A1:B3

fx

Vehicle

A

1

Vehicle\_type

AVERAGE

2

Car

3

Passenger

4

Grand Total

27

5



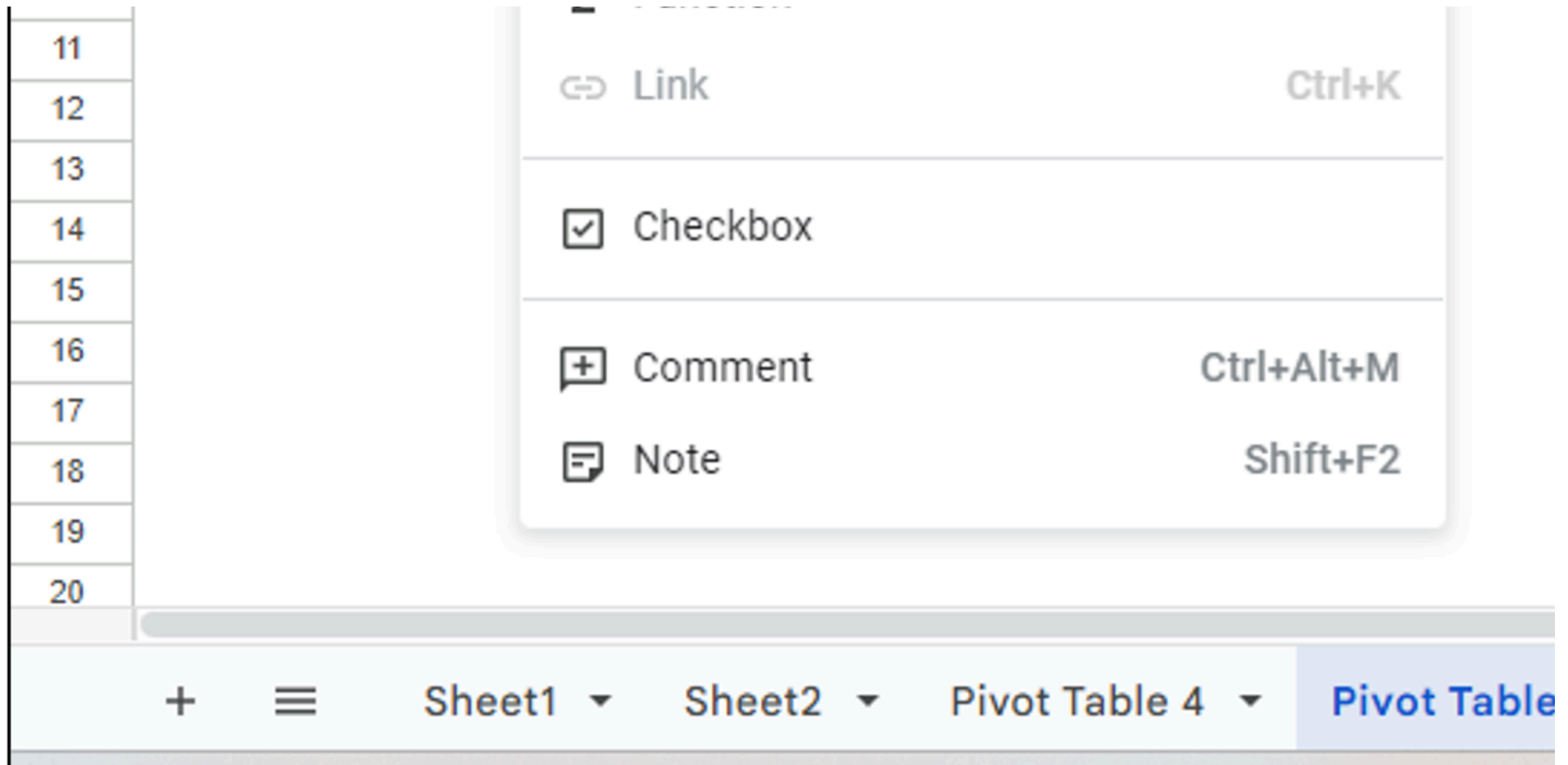
6

7

8

9

10



4. Select **Pie chart** under the **Chart type**, and your chart is ready.



## Car sales data



File Edit View Insert Format Data Tools Extensions Help



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Defaul...



A1:B3

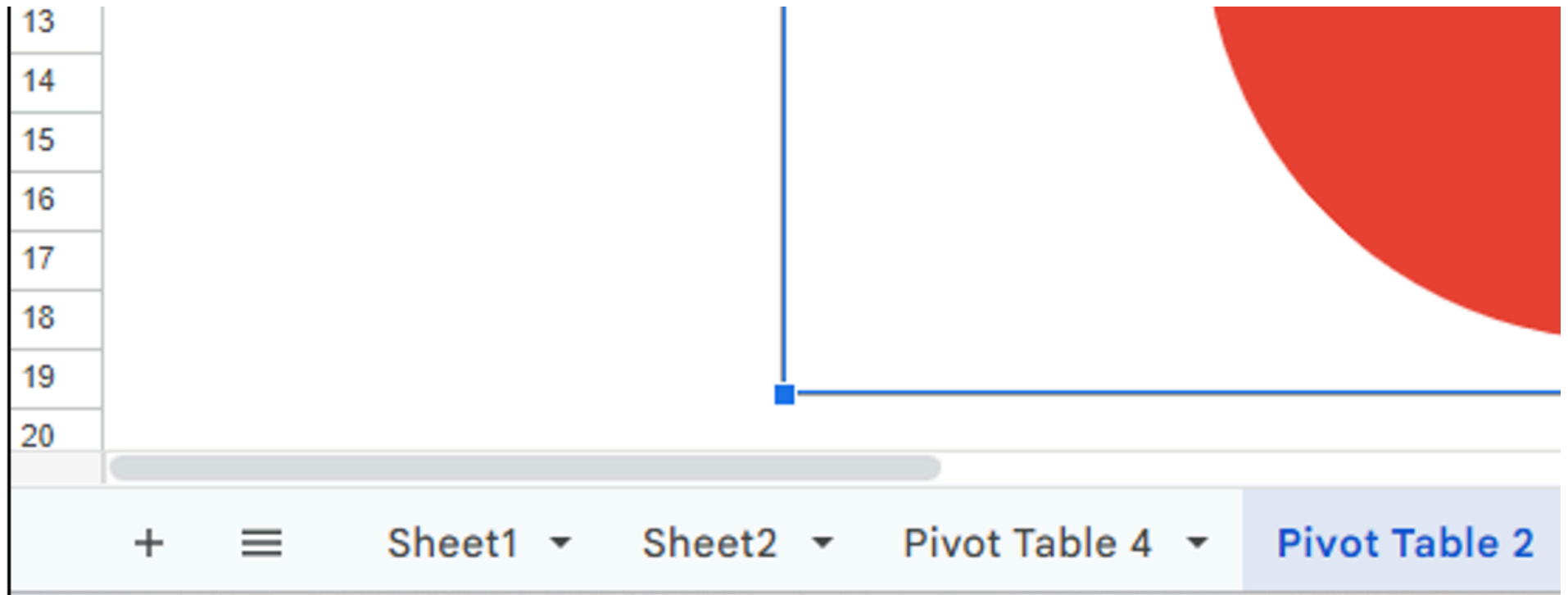


fx Vehicle\_type

|    | A            | B              | C | D | E |
|----|--------------|----------------|---|---|---|
| 1  | Vehicle_type | AVERAGE of Pri |   |   |   |
| 2  | Car          | 26.319975      |   |   |   |
| 3  | Passenger    | 27.7632        |   |   |   |
| 4  | Grand Total  | 27.39075484    |   |   |   |
| 5  |              |                |   |   |   |
| 6  |              |                |   |   |   |
| 7  |              |                |   |   |   |
| 8  |              |                |   |   |   |
| 9  |              |                |   |   |   |
| 10 |              |                |   |   |   |
| 11 |              |                |   |   |   |
| 12 |              |                |   |   |   |

AVERAGE of Price\_in\_thous

Passenger  
51.3%



5. To create the scatter plot for engine size vs. horsepower, let's get help from ChatGPT. Continuing with the same ChatGPT session, enter the following prompt.

How to create engine size vs. horsepower scatter plot for Google spreadsheet data?



You

How to create engine size vs horsepower scatter plot for google spreadsheet data?



ChatGPT

Creating a scatter plot for Engine Size vs. Horsepower in Google Sheets involves a few simple steps. Here's a detailed guide:

### Step 1: Open Your Google Spreadsheet

Open the Google Spreadsheet containing your car sales data.

### Step 2: Select Data

Assuming your Engine Size is in column H and Horsepower is in column I, follow these steps:

1. Click and drag to select the data in column  and I.

### Step 3: Insert Scatter Plot

Now, insert a scatter plot:

1. Go to **'Insert'** in the top menu.
2. Choose **'Chart'**.

### Step 4: Configure Scatter Plot

In the Chart Editor on the right:

1. **Chart Type:**
  - Choose "Scatter chart."
2. **Axis and Series:**
  - Assign "Horsepower" to the Y-axis.
  - Assign "Engine Size" to the X-axis.
3. **Customize:**
  - Customize the chart title, axis labels, and other options as needed.
4. Click "Insert" to add the scatter plot to your  eet.

6. To create the scatter plot for the columns "Engine\_size" and "Horsepower". go to the sheet **Pivot Table 3** and highlight the pivot table data. Click on **Insert** and then click on **Charts**.



Car sales data



File Edit View **Insert** Format Data Tools Ex



Cells



Rows



Columns



Sheet



**Chart**



Pivot table

A1:B33



fx

Engine

A

1

Engine\_size

AVER

2

#

3

1

4

1.5

5

1.6

|    |     |     |                                                                                               |
|----|-----|-----|-----------------------------------------------------------------------------------------------|
| 6  | 1.8 |     |  Image      |
| 7  | 1.9 |     |                                                                                               |
| 8  | 2   | 12  |  Drawing   |
| 9  | 2.2 |     |                                                                                               |
| 10 | 2.3 |     | $\Sigma$ Function                                                                             |
| 11 | 2.4 | 15  |                                                                                               |
| 12 | 2.5 | 15  |  Link      |
| 13 | 2.7 |     |                                                                                               |
| 14 | 2.8 | 19  | <input checked="" type="checkbox"/> Checkbox                                                  |
| 15 | 2.9 |     |                                                                                               |
| 16 | 3   | 19  |  Comment |
| 17 | 3.1 |     |                                                                                               |
| 18 | 3.2 | 21  |  Note    |
| 19 | 3.3 | 170 |                                                                                               |
| 20 | 3.4 | 215 |                                                                                               |

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Sheet1 ▼

Sheet2 ▼

Pivot Table ▼

7. Select **Scatter chart** under the **Chart type**, and your scatter plot is ready.



Car sales data



File Edit View Insert Format Data Tools Extensions Help



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.00

123

Default...

A1:B33



fx Engine\_size

|    | A           | B             | C | D | E |
|----|-------------|---------------|---|---|---|
| 1  | Engine_size | AVERAGE of Ho |   |   |   |
| 2  |             | #DIV/0!       |   |   |   |
| 3  | 1           | 55            |   |   |   |
| 4  | 1.5         | 92            |   |   |   |
| 5  | 1.6         | 106           |   |   |   |
| 6  | 1.8         | 132.375       |   |   |   |
| 7  | 1.9         | 128.8         |   |   |   |
| 8  | 2           | 127.5294118   |   |   |   |
| 9  | 2.2         | 130.5         |   |   |   |
| 10 | 2.3         | 183.5         |   |   |   |
| 11 | 2.4         | 152.7272727   |   |   |   |

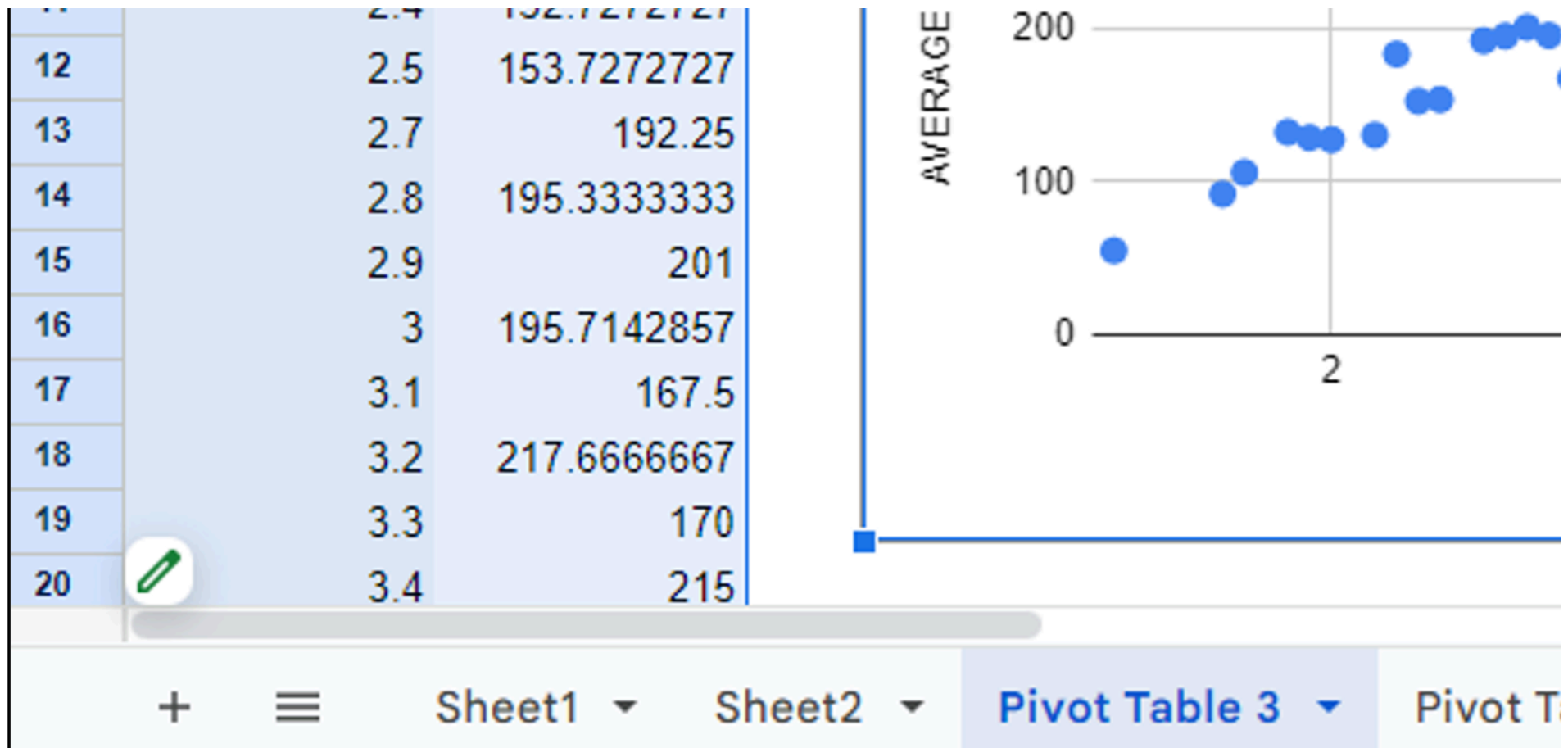
AVERAGE of Horsepower

500

400

300

of Horsepower



8. You can click the **Customize** button to access formatting options like changing the chart name, axis titles, and so on.



Car sales data



File Edit View Insert Format Data Tools Extensions



100%

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.0

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123

Default

A1:B33



fx Engine\_size

|    | A           | B             | C | D |
|----|-------------|---------------|---|---|
| 1  | Engine_size | AVERAGE of Ho |   |   |
| 2  |             | #DIV/0!       |   |   |
| 3  | 1           | 55            |   |   |
| 4  | 1.5         | 92            |   |   |
| 5  | 1.6         | 106           |   |   |
| 6  | 1.8         | 132.375       |   |   |
| 7  | 1.9         | 128.8         |   |   |
| 8  | 2           | 127.5294118   |   |   |
| 9  | 2.2         | 130.5         |   |   |
| 10 | 2.3         | 183.5         |   |   |

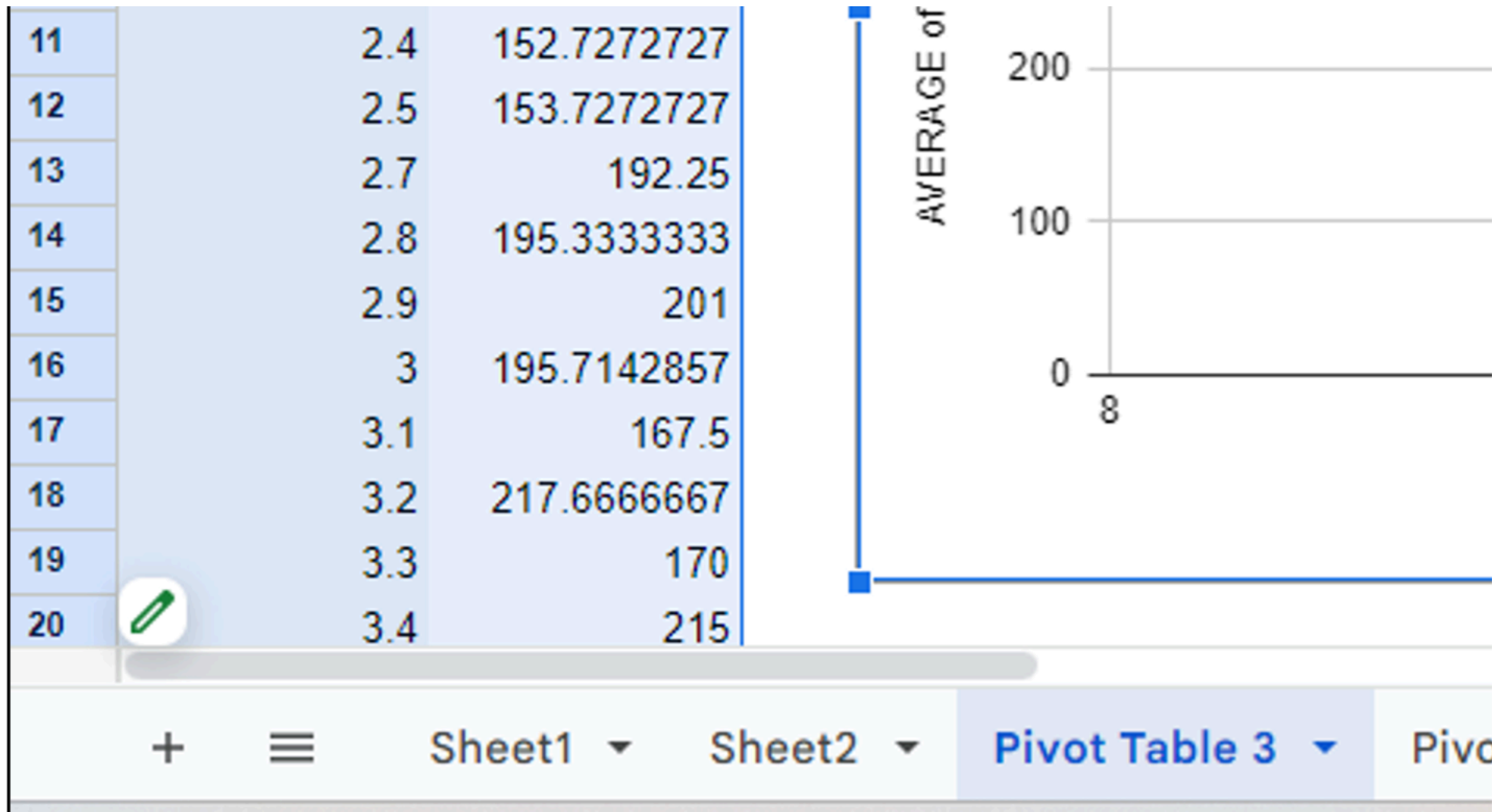
AVERAGE of Horsepower

500

400

300

Horsepower



9. To create a line chart for showing sales trends, let's get help from ChatGPT. In the same ChatGPT session, enter the following prompt.

How to create a line chart on Google spreadsheet using the pivot table data for 'Manufacturer, Model and Sales\_in\_thousand'?



**You**

How to create a line chart on Google spreadsheet using the pivot table data for 'Manufacturer, Model and Sales\_in\_thousand'?



**ChatGPT**

Creating a line chart in Google Sheets based on pivot table data for 'Manufacturer, Model, and Sales\_in\_thousand' involves a few steps. Here's a detailed guide:

### Step 1: Create a Pivot Table

**1. Select Your Data:**

- Highlight the data range, including headers, for Manufacturer, Model, and Sales\_in\_thousand.

**2. Insert Pivot Table:**

- Go to **'Data'** in the top menu.
- Choose **'Pivot table'**.

**3. Configure Pivot Table:**

- In the Pivot table editor that appears on the right:
  - Drag 'Manufacturer' to Rows.
  - Drag 'Model' to Rows, placing it below 'Manufacturer'.
  - Drag 'Sales\_in\_thousand' to Values, and set it to sum.

### Step 2: Create a Line Chart

**1. Select the Pivot Table Data:**

- Click on any cell in your pivot table.

**2. Insert Chart:**

- Go to **'Insert'** in the top menu.
- Choose **'Chart'**.

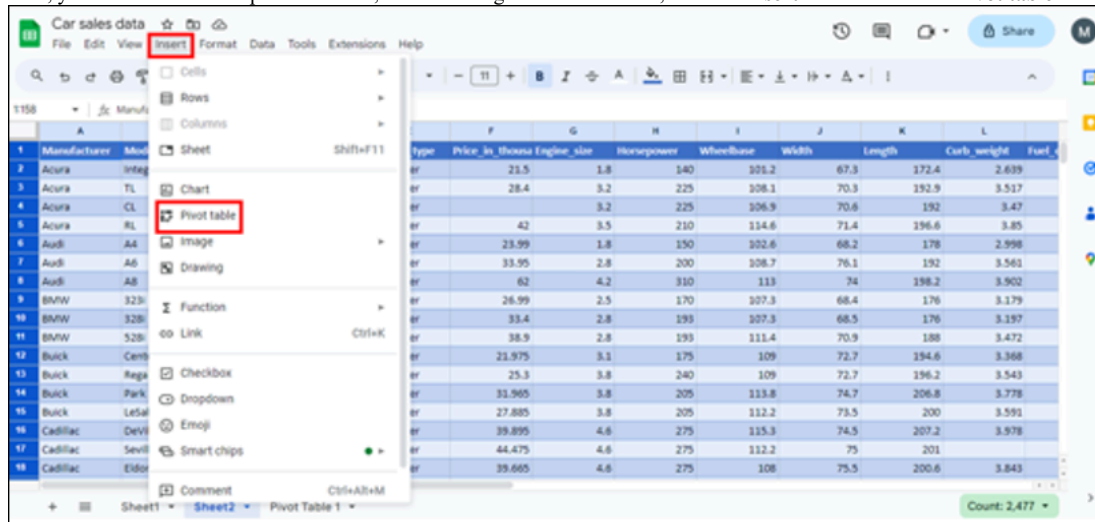
**3. Configure Chart:**

- In the Chart Editor:
  - Set 'Chart type' to "Line chart."
  - For 'X-axis,' choose 'Model.'
  - For 'Series,' choose 'Manufacturer.'
  - Make sure 'Values' is set to 'Sum of Sales\_in\_thousand.'

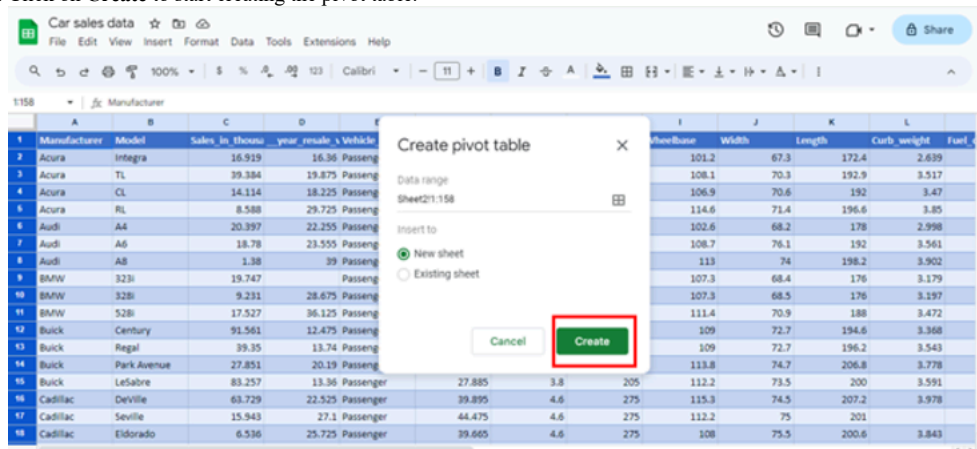
**4. Click "Insert" to add the line chart to your sheet.**



10. First, you need to create the pivot table. So, after selecting the entire data set, click on **Insert** and then click on **Pivot table**.



11. Click on **Create** to start creating the pivot table.



12. Drag **Manufacturer** and **Model** to **Rows** and **Sales\_in\_thousands** to the **Values** field.



Car sales data



File Edit View Insert Format Data Tools Extensions Help



100% ▾



123

Default... ▾

A1 ▾ | fx Manufacturer

|    | A            | B       | C               | D | E |
|----|--------------|---------|-----------------|---|---|
| 1  | Manufacturer | Model   | SUM of Sales_in |   |   |
| 2  | [-] Acura    | CL      | 14.114          |   |   |
| 3  |              | Integra | 16.919          |   |   |
| 4  |              | RL      | 8.588           |   |   |
| 5  |              | TL      | 39.384          |   |   |
| 6  | Acura Total  |         | 79.005          |   |   |
| 7  | [-] Audi     | A4      | 20.397          |   |   |
| 8  |              | A6      | 18.78           |   |   |
| 9  |              | A8      | 1.38            |   |   |
| 10 | Audi Total   |         | 40.557          |   |   |
| 11 | [-] BMW      | 323i    | 19.747          |   |   |
| 12 |              | 328i    | 9.721           |   |   |

|    |                                                                                            |             |         |
|----|--------------------------------------------------------------------------------------------|-------------|---------|
| 12 |                                                                                            | 520i        | 17.527  |
| 13 |                                                                                            | 528i        | 17.527  |
| 14 | BMW Total                                                                                  |             | 46.505  |
| 15 | - Buick                                                                                    | Century     | 91.561  |
| 16 |                                                                                            | LeSabre     | 83.257  |
| 17 |                                                                                            | Park Avenue | 27.851  |
| 18 |                                                                                            | Regal       | 39.35   |
| 19 | Buick Total                                                                                |             | 242.019 |
| 20 |  Cadillac | Catera      | 11.185  |



Sheet1 ▼

Sheet2 ▼

Pivot Table 5 ▼

Pivot Table



Car sales data



File Edit View Insert Format Data Tools Extensions Help



100%



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123

Default...

A1



fx

Manufacturer

|    | A            | B       | C               | D | E |
|----|--------------|---------|-----------------|---|---|
| 1  | Manufacturer | Model   | SUM of Sales_in |   |   |
| 2  | [-] Acura    | CL      | 14.114          |   |   |
| 3  |              | Integra | 16.919          |   |   |
| 4  |              | RL      | 8.588           |   |   |
| 5  |              | TL      | 39.384          |   |   |
| 6  | Acura Total  |         | 79.005          |   |   |
| 7  | [-] Audi     | A4      | 20.397          |   |   |
| 8  |              | A6      | 18.78           |   |   |
| 9  |              | A8      | 1.38            |   |   |
| 10 | Audi Total   |         | 40.557          |   |   |
| 11 | [-] BMW      | 323i    | 19.747          |   |   |

|                                                             |                                                                                            |             |         |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------|---------|
| 12                                                          |                                                                                            | 328i        | 9.231   |
| 13                                                          |                                                                                            | 528i        | 17.527  |
| 14                                                          | BMW Total                                                                                  |             | 46.505  |
| 15                                                          | – Buick                                                                                    | Century     | 91.561  |
| 16                                                          |                                                                                            | LeSabre     | 83.257  |
| 17                                                          |                                                                                            | Park Avenue | 27.851  |
| 18                                                          |                                                                                            | Regal       | 39.35   |
| 19                                                          | Buick Total                                                                                |             | 242.019 |
| 20                                                          |  Cadillac | Catera      | 11.185  |
|                                                             |                                                                                            |             |         |
| <div> + ≡ Sheet1 ▾ Sheet2 ▾ Pivot Table 5 ▾ Pivot Ta </div> |                                                                                            |             |         |

13. Click on the arrow under the **Values** field to show the SUM or COUNT as per your preference.



## Car sales data



File Edit View Insert Format Data Tools Extensions Help



100%



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%

.0



.00

123

Default...

A1



fx

Manufacturer

|    | A            | B       | C               | D | E |
|----|--------------|---------|-----------------|---|---|
| 1  | Manufacturer | Model   | SUM of Sales_in |   |   |
| 2  | [-] Acura    | CL      | 14.114          |   |   |
| 3  |              | Integra | 16.919          |   |   |
| 4  |              | RL      | 8.588           |   |   |
| 5  |              | TL      | 39.384          |   |   |
| 6  | Acura Total  |         | 79.005          |   |   |
| 7  | [-] Audi     | A4      | 20.397          |   |   |
| 8  |              | A6      | 18.78           |   |   |
| 9  |              | A8      | 1.38            |   |   |
| 10 | Audi Total   |         | 40.557          |   |   |
| 11 | [-] BMW      | 323i    | 19.747          |   |   |

|                                                                    |                                                                                            |             |         |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------|---------|
| 12                                                                 |                                                                                            | 328i        | 9.231   |
| 13                                                                 |                                                                                            | 528i        | 17.527  |
| 14                                                                 | BMW Total                                                                                  |             | 46.505  |
| 15                                                                 | – Buick                                                                                    | Century     | 91.561  |
| 16                                                                 |                                                                                            | LeSabre     | 83.257  |
| 17                                                                 |                                                                                            | Park Avenue | 27.851  |
| 18                                                                 |                                                                                            | Regal       | 39.35   |
| 19                                                                 | Buick Total                                                                                |             | 242.019 |
| 20                                                                 |  Cadillac | Catera      | 11.185  |
|                                                                    |                                                                                            |             |         |
| <div> + ≡ Sheet1 ▾ Sheet2 ▾ Pivot Table 5 ▾ Pivot Table 6 ▾ </div> |                                                                                            |             |         |

14. The pivot table is ready. Now highlight the pivot table data, click on **Insert**, and then click **Charts**.



Car sales data



File

Edit

View

Insert

Format

Data

Tools

Ext



Cells



Rows



Columns



Sheet



Chart



Pivot table



Image



Drawing

A1:C188



fx

Manufa

A

1

Manufacturer

Model

2

Acura

CL

3

Integr

4

RL

5

TL

6

Acura Total

7

Audi

A4

|    |             |             |
|----|-------------|-------------|
| 8  |             | A6          |
| 9  |             | A8          |
| 10 | Audi Total  |             |
| 11 | BMW         | 323i        |
| 12 |             | 328i        |
| 13 |             | 528i        |
| 14 | BMW Total   |             |
| 15 | Buick       | Centra      |
| 16 |             | LeSabre     |
| 17 |             | Park Avenue |
| 18 |             | Regal       |
| 19 | Buick Total | 242.015     |
| 20 | Cadillac    | Catera      |
|    |             | 11.185      |

Drawing
 

---

 Function
 

---

 Link
 

---

☒ Checkbox
 

---

 Comment
 

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 Note

Sheet1
Sheet2
Pivot Table 5

15. Select **Line chart** under the **Chart** type. You can also select the type of line chart you prefer.



# Car sales data



File Edit View Insert Format Data Tools Extensions



100%



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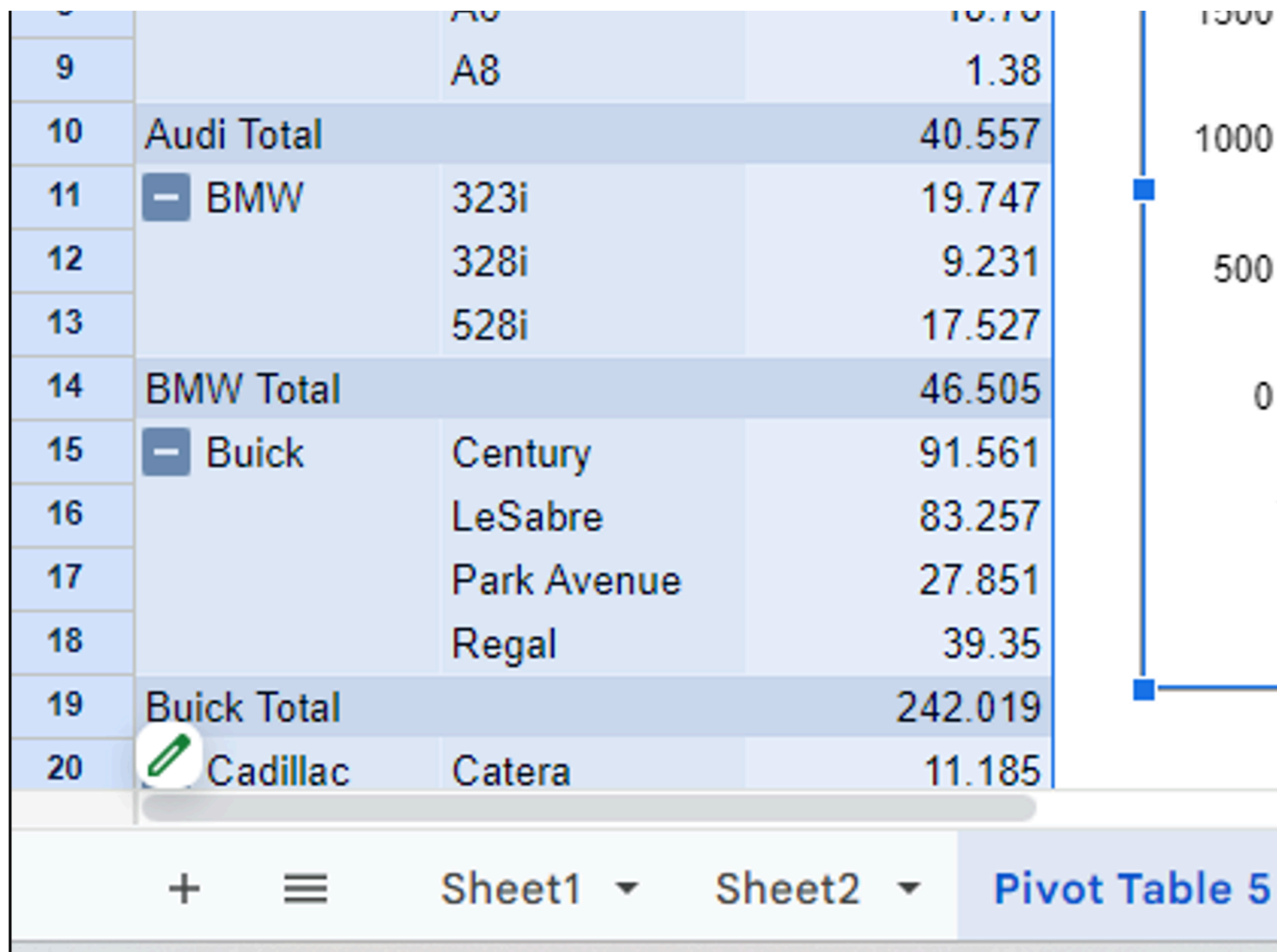
123

A1:C188



fx Manufacturer

|   | A            | B       | C               | D    |
|---|--------------|---------|-----------------|------|
| 1 | Manufacturer | Model   | SUM of Sales_in |      |
| 2 | - Acura      | CL      | 14.114          |      |
| 3 |              | Integra | 16.919          |      |
| 4 |              | RL      | 8.588           | 2500 |
| 5 |              | TL      | 39.384          |      |
| 6 | Acura Total  |         | 79.005          | 2000 |
| 7 | - Audi       | A4      | 20.397          |      |
| 8 |              | A6      | 18.78           | 1500 |

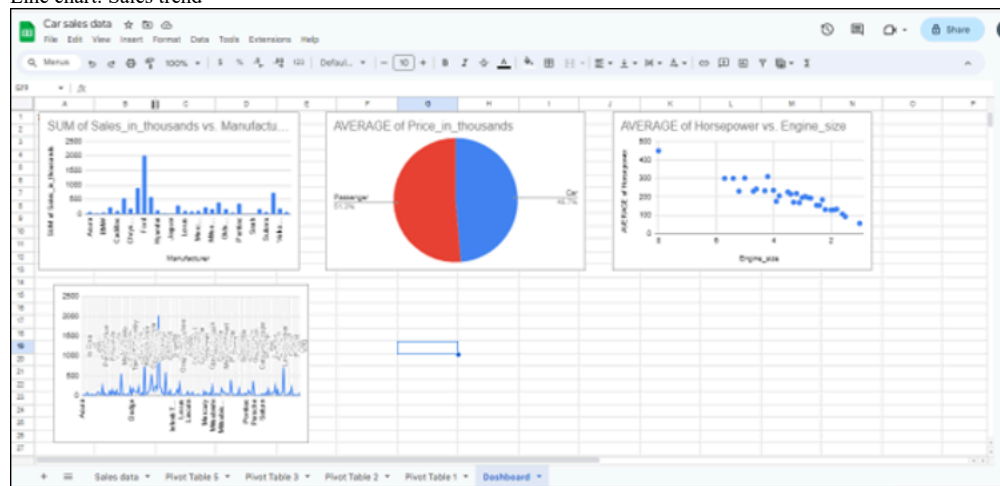


Step 5: Assemble the dashboard

As the final step, you need to assemble all the elements in a sales dashboard.

1. To arrange the charts, place each of them on a dashboard sheet and adjust the size and position for a clean layout.
2. The dashboard created using ChatGPT and Google spreadsheet is shown in the following image, showcasing the following charts:

- Bar chart: Sales by manufacturer
- Pie chart: Average sales
- Scatter plot: Eng. Size vs. Horsepower
- Line chart: Sales trend



## Summary

Congratulations! You just completed the hands-on lab Generative AI Applications for Managers.

In this lab, you explored how to use generative AI tools to create dashboards to analyze performance and get insights and recommendations.

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# Skills Network